

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 22: Trigonometrical Ratios

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In triangle ABC, $\angle B = 90^\circ$, $AB = 8$ cm, $BC = 6$ cm. Find $\sin A$ and $\cos A$.
2. If $\sin \theta = \frac{5}{13}$, find $\cos \theta$ and $\tan \theta$.
3. If $\tan A = \frac{3}{4}$, find $\sin A + \cos A$.
4. If $\sec \theta = \frac{25}{7}$, find $\tan \theta$ and $\operatorname{cosec} \theta$.
5. If $\cos \theta = \frac{3}{5}$, find $\frac{\sin \theta - \cot \theta}{2 \tan \theta}$.
6. If $\tan \theta = \frac{5}{12}$, find $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$.
7. If $3 \tan \theta = 4$, find $\frac{4 \cos \theta - \sin \theta}{2 \cos \theta + \sin \theta}$.
8. In triangle PQR, $\angle Q = 90^\circ$, $PR = 26$ cm and $\cos P = \frac{5}{13}$. Find PQ and QR.
9. If $\sin \theta = \cos \theta$ for an acute angle, find $\tan \theta$.
10. Is $\sin \theta = \frac{4}{3}$ possible for an acute angle? Why?

Answer Key

1. $AC = 10$ cm; $\sin A = \frac{3}{5}$, $\cos A = \frac{4}{5}$

2. $\frac{12}{13}$ and $\frac{5}{12}$

3. $\frac{7}{5}$

4. $\frac{24}{7}$ and $\frac{25}{24}$

5. $\frac{\frac{4}{5} - \frac{3}{4}}{\frac{8}{3}} = \frac{3}{160}$

6. $-\frac{17}{7}$

7. $\frac{4 - \frac{4}{3}}{2 + \frac{4}{3}} = \frac{4}{5}$

8. 10 cm and 24 cm

9. 1

10. No; the perpendicular cannot be longer than the hypotenuse